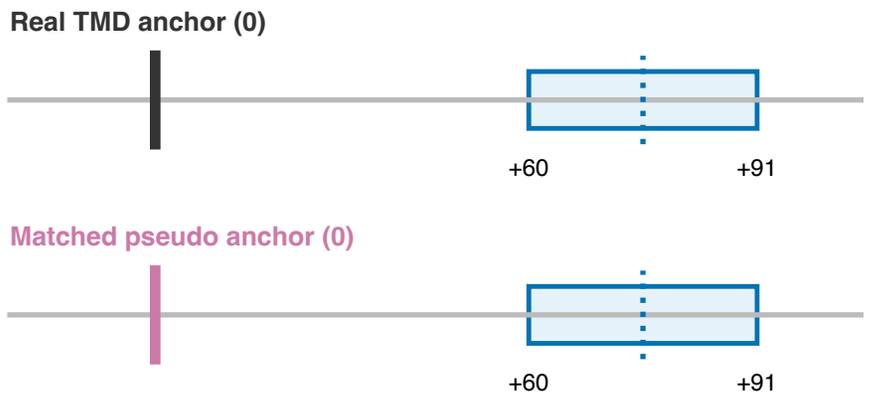
TMD-relative ribosome profiles: estimand, controls and hierarchy

A

Equivalent real and pseudo estimands



$D = \log_2(\text{real pulldown/input}) - \log_2(\text{pseudo pulldown/input})$

B

Falsification before interpretation

Apparent TMD-relative profile

↓

Matched pseudo anchors

↓

Coordinate null

↓

Deduplication / read-length sensitivity

↓

Calibrated biological interpretation

C

Frozen evidence hierarchy

PRESPECIFIED PRIMARY
+75 nt universal TMD-specific enrichment

PRESPECIFIED SECONDARY
11-center profiles; FIRST vs INTERNAL

POST-REVEAL
Common anchors; multi-pass; direct interaction

POST-REVEAL REGION
+90/+105/+120 composite; +60…+120 sensitivity

Later analyses do not replace the negative primary result.

D

Interpretation boundary

Permitted
post-reveal, replicate-concordant association
internally stable under specified sensitivities

evidence boundary

↓

Not established
targeting • insertion • causality • universal mechanism
validation • independent replication